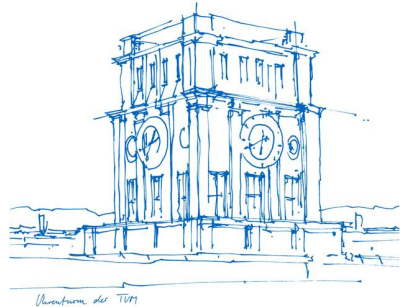


FineFlow–Espresso: A Calibratable 1D Model of Fines Migration

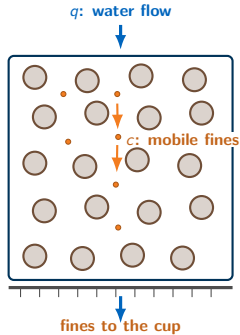
**A continuum model linking fines transport
and hydraulic resistance**

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Flow and fines inside the coffee bed



Flow and mobile fines

- Q : flow rate; A : puck area; $q = Q/A$: superficial velocity;
- c : mobile-fines concentration; $q c$: advective fines flux.

Pore structure and retained fines

- ε : porosity; K : permeability;
- m_a : remaining initially attached fines;
 m_d : recaptured deposits;
- m_a, m_d : mass per bulk volume; c : mass per water volume.

Distinct retained populations assume different kinetics or hydraulic effects.

Scope of the 1D continuum model

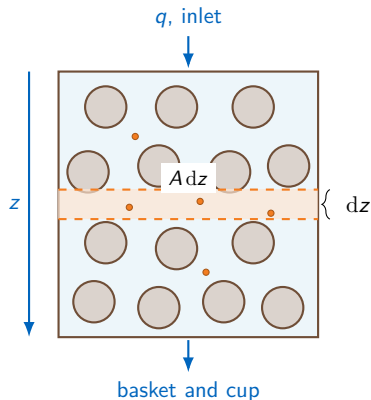
Present approach

- Forward flow with cross-section-averaged fields varying in z and t .
- CT stage snapshots can constrain axial structural changes.
- Pressure and flow histories constrain the evolving total resistance.

Limits and next measurements

- **Reverse flow and radial channeling are outside the scope of this work.**
- CT shows radial heterogeneity. Measurements must test whether axial averaging captures the overall hydraulic response.
- Time-resolved cup-fines data are a proposed measurement; availability is unconfirmed.
- Pore networks and 2D/3D continuum models are future extensions.

Continuum balance over an infinitesimal slice



Water-filled pore space

$$dM_m = \varepsilon c A dz$$

Mobile fines cross the slice boundaries by advection and dispersion.

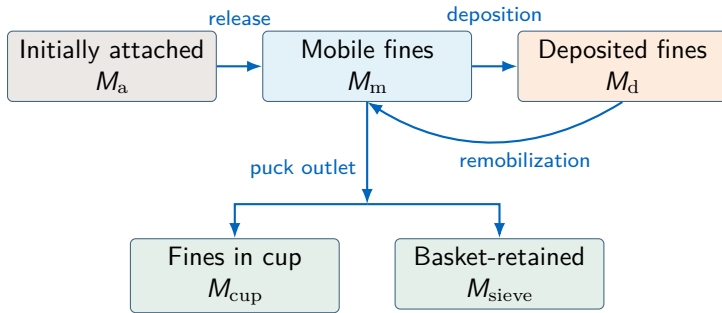
Stationary coffee skeleton

$$dM_a = m_a A dz, \quad dM_d = m_d A dz$$

Release and capture exchange fines locally between water and skeleton.

The highlighted slice defines the continuum balance. It is not a numerical cell.

Fines exchange locally and leave through the outlet



$$M_0 = M_a + M_m + M_d + M_{sieve} + M_{cup}.$$

$$M_a = A \int m_a dz, M_m = A \int \varepsilon c dz, M_d = A \int m_d dz.$$

Accounting domain: puck + basket + collected cup fines.

Continuum equations and optional surface loading

$$\frac{\partial(\varepsilon c)}{\partial t} + \frac{\partial(qc)}{\partial z} = \frac{\partial}{\partial z} \left(\varepsilon D \frac{\partial c}{\partial z} \right) + R_{\text{rel}} - R_{\text{dep}} + R_{\text{wash}},$$

$$\frac{\partial m_a}{\partial t} = -R_{\text{rel}}, \quad \frac{\partial m_d}{\partial t} = R_{\text{dep}} - R_{\text{wash}}.$$

Equivalent surface description

$$m_a = \sigma c_a, \quad m_d = \sigma c_d, \quad [\sigma] = \text{m}^2/\text{m}_{\text{bulk}}^3, \quad [c_a] = [c_d] = \text{kg}/\text{m}^2.$$

Use $\partial_t(\sigma c_a)$ if the specific surface area changes.

Only mobile fines travel axially. All R terms have units $\text{kg}/(\text{m}_{\text{bulk}}^3 \text{ s})$.

Release and capture are constitutive hypotheses

Release of initially attached fines

$$R_{\text{rel}} = k_{\text{rel}} m_a (|q|/q_{\text{ref}})^{a_r}, \quad [k_{\text{rel}}] = \text{s}^{-1}.$$

Capture of mobile fines

$$R_{\text{dep}} = k_{\text{dep}} \chi(z) |q| c \left[1 - \frac{m_d}{m_{\text{cap}}} \right]_+, \quad [k_{\text{dep}}] = \text{m}^{-1}.$$

- Release depends on the remaining available inventory and flow.
- Capture depends on mobile flux and available deposit capacity.
- $\chi = 1$ is the unbiased baseline. Axial weighting needs evidence.
- Compare these laws using multiple flow rates and spatial deposit data.

Deposit remobilization and local pore accessibility

$$R_{\text{det}}^* = k_{\text{det}} m_d \left[\frac{G}{G_{\text{crit}}} - 1 \right]_+^b, \quad G = \left| \frac{\partial p}{\partial z} \right|.$$

$$R_{\text{wash}} = P_{\text{esc}} R_{\text{det}}^*, \quad P_{\text{esc}} = \frac{1}{1 + \exp[s(d_f/d_h - \lambda_c)]}.$$

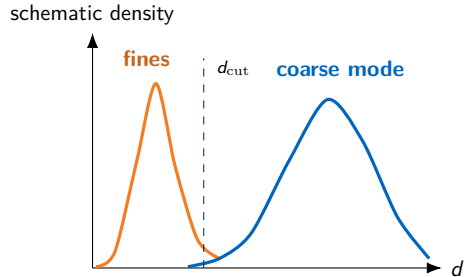
- P_{esc} is an effective fraction entering the mobile phase locally.
- Unsuccessful mobilization remains in m_d in this coarse description.
- A mobile particle can be captured again before reaching the outlet.
- The logistic law is unvalidated. Compare with $P_{\text{esc}} = 1$.

Local remobilization does not imply arrival in the cup.

The PSD is reduced to quantities used by the 1D closures

Measured bimodal PSD

- Fit a five-parameter mixture model.
- Choose a mobility cutoff d_{cut} .
- Estimate the fine fraction and its releasable share.
- Calculate representative fine and coarse diameters.



$$f_{\text{fine}} + \text{releasable share} \rightarrow m_{a,0}, \quad d_f \rightarrow d_f/d_h, \quad d_{\text{eff}} \rightarrow d_{h,0} \text{ and structural baseline.}$$

Source: Bora and Briesen, Journal of Food Process Engineering 49 (2026), e70645.

Two alternative closures for the same permeability

Both closures supply K to Darcy's law. They describe how deposits change resistance.

$$\varepsilon = \max(\varepsilon_0 - m_d/\rho_d, \varepsilon_{\min})$$

Kozeny–Carman: structural baseline

$$\frac{K}{K_0} = \left(\frac{\varepsilon}{\varepsilon_0}\right)^3 \left(\frac{1 - \varepsilon_0}{1 - \varepsilon}\right)^2$$

Relates permeability to porosity under fixed effective particle scale and Kozeny factor.

This simplified relation may miss changes in pore connectivity and localized plugging.

Exponential: empirical comparison

$$\frac{K}{K_0} = \exp\left(-\beta \frac{m_d}{m_{\text{cap}}}\right)$$

Assumes a direct sensitivity to deposit loading. β controls the imposed blockage strength.

Its form and coefficient were not derived from coffee measurements in this study.

The runs test closure sensitivity at identical K_0 . They do not validate either law.

Why the illustrative model used an exponential law

Motivation: represent strong blockage with a simple adjustable law

The hypothesis was that fines obstruct narrow flow paths, allowing a large resistance increase with a modest change in total porosity.

The exponential form stays positive and decreases smoothly. These useful mathematical properties do not establish physical validity.

m_d/m_{cap}	K/K_0 at $\beta = 5.5$
0	1
0.5	0.064
1	0.0041

An illustrative assumption

Neither the exponential law nor $\beta = 5.5$ was fitted to the coffee data or taken from a cited coffee study. Values shown are before any numerical lower bound.

Strong blockage is substantially imposed by this closure; it is not independent evidence of the mechanism.

Darcy flow and the permeability closure have different roles

Flow law: pressure gradient for a given superficial velocity

$$-\frac{\partial p}{\partial z} = \frac{\mu q}{K(z, t)}.$$

Structural closure: permeability of the evolving puck

$$K(z, t) = K_0(z) f_{\text{KC}}(\varepsilon(z, t)).$$

Total hydraulic resistance

$$\Delta p_{\text{puck}} = q \int_0^L \frac{\mu}{K(z, t)} dz.$$

Darcy and Kozeny–Carman are complementary. Forchheimer adds an optional inertial term to Darcy.

Darcy gives practically the same results

We ran the same case twice, changing only the flow law.

Result at 60 s	Darcy–Forchheimer	Darcy
Pressure [bar]	8.9906	8.9906
Flow rate [mL/s]	0.04535	0.04535
Cup fines [g]	0.03993	0.03993

Values rounded for display.

Across the entire simulation, the largest pressure difference was only **0.0203 Pa**.

We therefore use Darcy for this tested case.

Controlled 60 s simulations with identical exponential permeability, kinetics and PI settings.

Operating conditions and basket resistance

$$\Delta p_{\text{total}} = \Delta p_{\text{puck}} + QR_s, \quad \dot{M}_{\text{cup}} = (1 - \eta_s) A q c(L, t).$$

- R_s describes basket pressure loss. η_s describes fines retention.
- Prescribed pressure or flow is sufficient for controlled boundary conditions.
- PI mode reproduces a machine response only after characterizing that machine.

$$e = p_{\text{set}} - p, \quad Q_{\text{cmd}} = \text{clip} \left(Q_{\text{ff}} + K_P e + K_I \int e dt, 0, Q_{\text{max}} \right).$$

PI has no derivative term. Identify K_P , K_I , pump lag and flow cap independently.

Boundary conditions for clean feed and an open outlet

$$J = qc - \varepsilon D \partial_z c$$

Inlet: no fines supplied

$$J(0, t) = qc_{\text{in}} = 0$$

The feed is clean water. We prescribe the total incoming fines flux.

This does not require imposing both $c = 0$ and $\partial_z c = 0$ on the interior solution.

In the code, boundary dispersive fluxes are omitted; inlet total flux is zero and outlet flux is qc . Reconsider this closure if downstream mixing or back-dispersion matters.

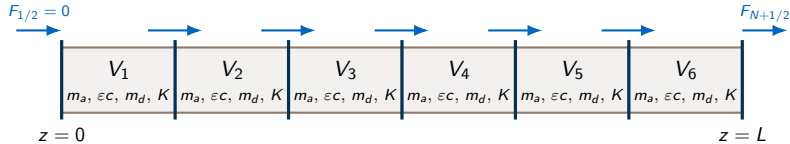
Outlet: convective removal

$$\partial_z c(L, t) = 0, \quad J(L, t) = qc(L, t)$$

We neglect dispersive exchange with the downstream basket/cup region.

This open-outlet approximation allows fines to leave by advection, without a modeled downstream concentration field.

The finite-volume method enforces flux balance cell by cell



$$\frac{(\varepsilon c)_i^{n+1} - (\varepsilon c)_i^n}{\Delta t} = -\frac{F_{i+1/2} - F_{i-1/2}}{\Delta z} + R_{\text{rel},i} - R_{\text{dep},i} + R_{\text{wash},i}.$$

$$F_{i+1/2} = \underbrace{q c_i}_{\text{advection}} - \underbrace{\varepsilon_{i+1/2} D \frac{c_{i+1} - c_i}{\Delta z}}_{\text{dispersion}}, \quad q \geq 0.$$

First-order upwind advection and centred dispersion. No high-order flux limiter.

Shared face fluxes cancel between adjacent cells, preserving the global mass balance.

The smallest timescale limits the numerical step

$$\Delta t = \min(\Delta t_{\max}, \Delta t_{\text{transport}}, \Delta t_{\text{pump}}, t_{\text{end}} - t)$$

$$\Delta t_{\text{transport}} = \frac{0.42}{|q|/(\varepsilon_{\min} \Delta z) + 2D/\Delta z^2}$$

Limit	Purpose
$\Delta t_{\max}$	User-selected upper bound for temporal resolution.
$\Delta t_{\text{transport}}$	Combined explicit advection–dispersion restriction. Here $\varepsilon_{\min} = \min_i \varepsilon_i(t)$.
$\Delta t_{\text{pump}} = \tau_p/4$	Used only in PI mode to resolve pump response; this is not the PI integral time.
$t_{\text{end}} - t$	Stops exactly at the requested final time.

For $D = 0$: $\text{Co} = |q|\Delta t/(\varepsilon_{\min} \Delta z) \leq 0.42$. Timestep refinement still checks accuracy of kinetics and coupling. The transport bound is the implemented estimate, not a general heterogeneous-field proof.

Why finite volumes for fines transport?

The main requirement is consistent mass accounting

- Each cell stores a fines inventory and exchanges mass through its faces.
- One shared face flux leaves one cell and enters the next.
- Summing the cells cancels internal fluxes; boundary fluxes give collected fines.
- Local release, capture and remobilization transfer mass between inventories.

Alternative	Assessment for this model
Finite differences	Equally viable in conservative flux form; on a uniform grid, they can give the same update.

FVM is a transparent first implementation. Its first-order upwind diffusion requires grid refinement.

Pressure follows the current state algebraically

Quasi-steady hydraulic approximation

$$q(t) = Q(t)/A, \quad p(z, t) = QR_s + q \int_z^L \frac{\mu}{K(\zeta, t)} d\zeta.$$

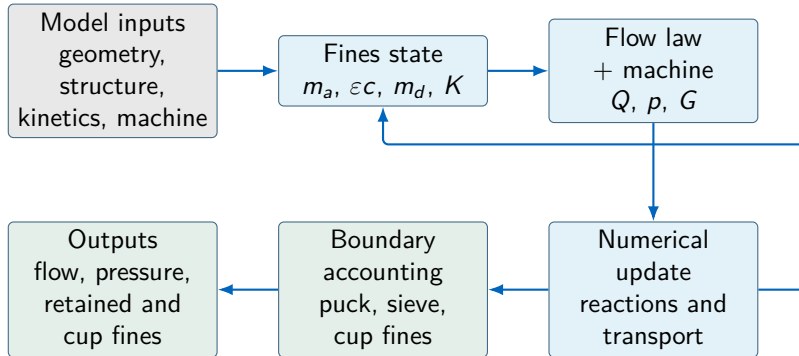
- Flow mode supplies $Q(t)$ directly.
- Pressure mode gives

$$Q = \frac{\Delta p_{\text{total}}}{R_s + (1/A) \int_0^L \mu/K dz}.$$

- PI mode evolves the pump state and uses the same resistance relation.

No additional fluid momentum state is integrated. Fines and machine states can still evolve in time.

Coupled physics and its numerical update



Physical coupling: deposit changes ε , K and d_h ; these change pressure and flow; the new hydraulics then alter release, deposition and washout.

Quasi-steady flow is a physical assumption. Update ordering is a numerical choice.

Calibration begins with independently measured structure

Input	Measurement or proposed inference
Geometry and fluid	Puck height and area, temperature-dependent viscosity and density.
PSD and available fines	Fine fraction, representative sizes and an independently constrained releasable share.
Initial structure	CT porosity and heterogeneity. Accessible specific surface σ with resolution uncertainty.
Hydraulics	Low-flow pressure–flow slope for K_0 , empty-basket loss for R_s .
Future pore networks	Connectivity-sensitive permeability and capture/accessibility closures.

CT-derived hydraulic diameter is a structural proxy, not a direct measurement of every pore throat.

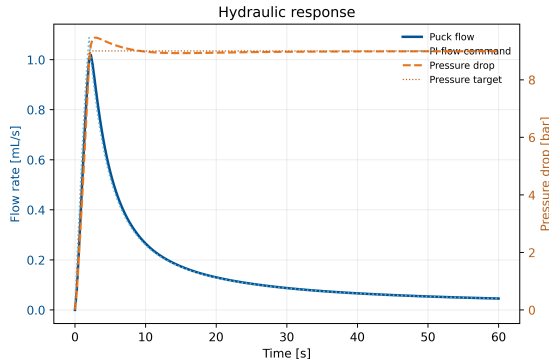
Experiments to distinguish competing kinetic models

- Fines collected in time intervals at several flows constrain release and capture.
- Interrupted CT locates retained material and redistribution.
- Pressure pulses test deposit remobilization and threshold behavior.
- Outlet size distributions test size selectivity.
- Basket-retained mass separates puck transport from cup collection.

Model comparisons One versus two retained populations; accessibility factor versus $P_{\text{esc}} = 1$; uniform capture versus axial weighting.

These are proposed experiments and comparisons. Validate on withheld pucks or grind settings.

The illustrative case develops strong viscous blockage



At 60 s

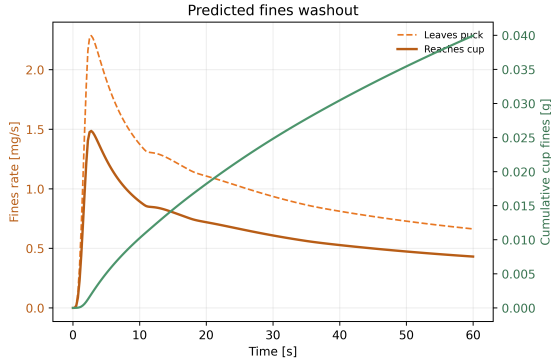
- total pressure: 8.99 bar;
- flow: 0.045 mL/s;
- minimum K/K_0 : 0.0126.

Interpretation

The PI controller approaches 9 bar while the increasing puck resistance strongly reduces flow.

New Darcy run: exponential permeability, PI control. Illustrative parameters, no experimental calibration.

The model distinguishes puck washout from fines in the cup



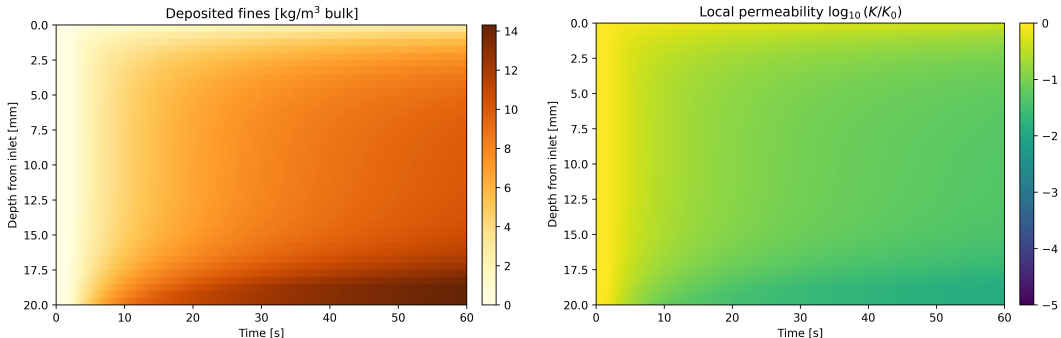
Cumulative masses

- puck outlet: 0.0614 g;
- basket retained: 0.0215 g;
- cup: 0.0399 g.

With $\eta_s = 0.35$, the basket retains 35% of the fines that leave the puck.

New Darcy run with exponential permeability; constant basket retention.

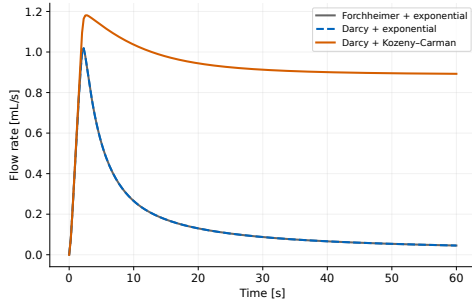
The axial fields locate the effective blocking region



These fields can be compared with interrupted CT reconstructions, but a 1D layer cannot represent lateral channel geometry.

New Darcy run with exponential permeability. Closure comparison follows.

Permeability closure strongly changes the flow response



At 60 s

Exponential:

$$Q = 0.0454 \text{ mL/s}$$

Kozeny–Carman:

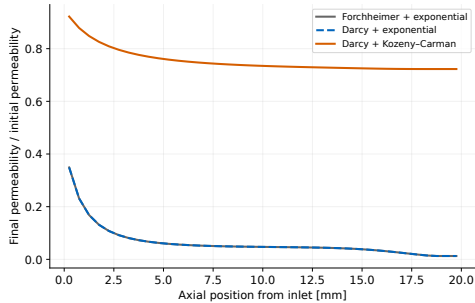
$$Q = 0.892 \text{ mL/s}$$

Both Darcy runs approach
the 9 bar PI target.

The closure changes resistance and flow, even with nearly the same final pressure.

Same initial K_0 , kinetics and machine parameters. Gray and blue curves overlap: inertia is negligible.

Kozeny–Carman predicts weaker permeability loss



Minimum K/K_0

Exponential:
0.0126

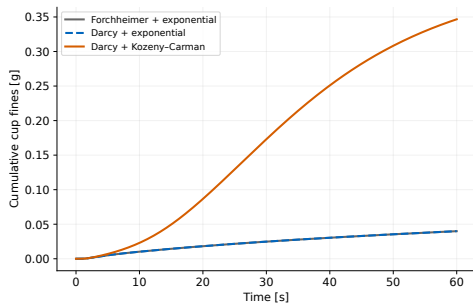
Kozeny–Carman:
0.723

The same deposit-to-porosity
law feeds both closures.

The exponential law imposes stronger blockage. The comparison does not establish which closure is correct.

Final profiles at 60 s. Coupled deposit fields also differ because flow feeds back on kinetics.

The permeability closure also changes cup fines



Cup fines at 60 s

Exponential:

0.0399 g

Kozeny-Carman:

0.3466 g

Same basket retention:

$\eta_s = 0.35$.

This is closure sensitivity, not validation. Neither prediction is an experimental target.

All kinetic and machine parameters held fixed. No fitting to reproduce the exponential-law result.

Numerical verification and physical validation

Verification

- Conservation and nonnegative inventories.
- Analytical hydraulic limiting cases.
- Grid and timestep convergence.
- Controller limits and boundary accounting.

Validation

- Pressure and flow measurements.
- CT structural changes.
- Collected fines and size distributions.
- Predictions for withheld cases.

Current unit tests: 15/15 passed. All three comparison runs conserve fines mass to relative error below 1.4×10^{-15} . Physical validation remains outstanding.

Next comparison: redistribution versus washout

Retained redistribution Fines release and move, but deposition elsewhere in the puck dominates. CT changes while cumulative outlet mass stays small.

Net washout Mobile fines reach the puck outlet. Basket retention and cup collection quantify the outgoing mass.

Proposed study

- Compare pressure or flow levels with the same initial puck structure.
- Record M_a , M_m , M_d and M_{out} with axial deposit profiles.
- Darcy is sufficient here. Compare both permeability predictions with data.
- Use PNM or 2D/3D only where pore connectivity or radial bypass demands it.

References and model documentation

- M. Bora and H. Briesen, “Characterization of Bimodal Particle Size Distribution of Ground Coffee Powder,” *Journal of Food Process Engineering*, 49, e70645, 2026.
doi:10.1111/jfpe.70645
- S. Ergun, “Fluid Flow through Packed Columns,” *Chemical Engineering Progress*, 48, 89–94, 1952.
- FineFlow–Espresso technical report, Python solver, named-case configuration and automated verification suite accompanying this presentation.